

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Mandatory Cross-Track Integration Report

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 6 Submission – Data Analytics Integration Evidence

Week 5 asked interns to identify at least one meaningful dependency or collaboration point between tracks. Week 6 requires that connection to actually be used: a completed cross-track integration activity that results in a meaningful change, improvement, decision, integration or validated output, not communication alone. This report documents the Data Science track's Week 6 integration with the Data Analytics track against that standard.

1. Track collaborated with

Data Analytics Track.

2. Project dependency

Both tracks are investigating the same HealthConnect objective — reducing missed appointments while improving, not degrading, the patient support experience — from complementary angles. The Data Analytics track identifies and statistically validates behavioural patterns in appointment attendance across the shared HealthConnect Appointment Dataset; the Data Science track depends on receiving genuine, non-arbitrary behavioural patterns of this kind to guide feature refinement, rather than engineering interaction terms speculatively. The Week 6 feature-refinement work (Model Improvement & Validation Report, Section 3) was explicitly dependent on this input, and the two engineered interaction features described in Section 5 below could not have been justified on the Data Science track's own analysis alone.

3. Information/output received

The Data Analytics track (Bada) shared, via Slack, the validated descriptive findings from its Week 5 analysis, extracted specifically to support the Data Science track's Week 6 feature refinement and error analysis (Figure 1). The findings covered: an overall no-show rate of 51.15% after excluding Cancelled appointments; previous no-show history as one of the strongest relationships (57.8% no-show rate for patients with at least one prior no-show, versus 46.3% with none), with comparatively small age-group differences (48.1%–53.0%); a strong increasing pattern by booking lead time (29.5% at 0–7 days, rising to 63.9% at 31–60 days); appointment-type differences (Follow-up highest at 54.2%, General Consultation lowest at 49.1%); comparatively modest appointment-time and day-of-week differences; a higher no-show rate where no reminder was sent (54.6%) than where one was (49.9%); a distance gradient (48.7% under 5 km, rising to 60.5% at 20+ km); and waiting-time groups found to be relatively uninformative by comparison. The message concluded with an explicit, actionable recommendation: to test interaction effects such as lead time × previous no-show history, lead time × reminder status, distance × lead time, and previous history × appointment type as candidate feature-refinement directions for the Data Science track.

Bada 2:40 PM

Hi, I extracted the main validated descriptive findings from my Data Analytics Week 5 analysis for your feature refinement and error analysis.

Overall baseline: After excluding Cancelled appointments, the eligible Attended/No-Show population had an overall no-show rate of 51.15%.

Patient characteristics/history:

Previous no-show history showed one of the strongest relationships. Appointments involving patients with at least one previous no-show recorded a 57.8% no-show rate compared with 46.3% among those with no previous no-show history. Age-group differences were much smaller: rates ranged from roughly 48.1% to 53.0%, with the 65+ group lowest at 48.1%.

Appointment characteristics:

Booking lead time showed a strong increasing pattern: 29.5% for 0–7 days, 35.2% for 8–14 days, 45.5% for 15–30 days and 63.9% for 31–60 days.

By appointment type, Follow-up appointments had the highest observed no-show rate (54.2%), followed by Diagnostic Test (52.0%), Specialist Consultation (50.5%) and General Consultation (49.1%).

Appointment-time differences were relatively modest: Evening 52.6%, Afternoon 51.2%, Morning 50.7%. Day-of-week differences were also relatively small, with Monday highest at 53.1% and Friday lowest at 48.7%.

Reminder patterns:

Appointments with no reminder recorded a 54.6% no-show rate compared with 49.9% where a reminder was sent. Reminder-channel analysis also suggested variation across channels, although I am treating this as descriptive rather than causal because reminder allocation may not be random.

Operational/access segments:

Distance showed a noticeable gradient: 48.7% for <5 km, 49.3% for 5–10 km, 52.3% for 10–20 km and 60.5% for 20+ km. Waiting-time groups were much more similar and therefore appeared less informative than lead time, previous history or distance.

Main takeaway:

The strongest Analytics signals are **previous no-show history, booking lead time and distance**, with reminder status and follow-up appointments as secondary areas worth investigating. For feature refinement/error analysis, I would particularly suggest testing interactions such as **lead time × previous no-show history, lead time × reminder status, distance × lead time, and previous history × appointment type**.

These are observational associations within the supplied dataset, so they should support modelling decisions rather than be interpreted as causal effects.

Figure 1. Slack message from the Data Analytics track (Bada) sharing validated Week 5 descriptive findings and specific interaction-testing suggestions.

4. Information/output provided

In return, the Data Science track shared its own Week 6 error-analysis findings with the Data Analytics track: the specific patient and appointment segments and conditions under which the Week 5 baseline model produces the most False Positives and False Negatives — for example, that shorter booking lead times and a cleaner prior attendance history are associated with missed no-shows (False Negatives), while longer lead times and a higher prior no-show count are associated with false alarms (False Positives) (Figure 2). This was supplemented with the resulting model-comparison results, coefficient and permutation-importance interpretation, and the provisional Week 6 candidate model and threshold requirements, so that the Data Analytics track could target further investigation at the model's current blind spots and independently check its own descriptive findings against separate model evidence.

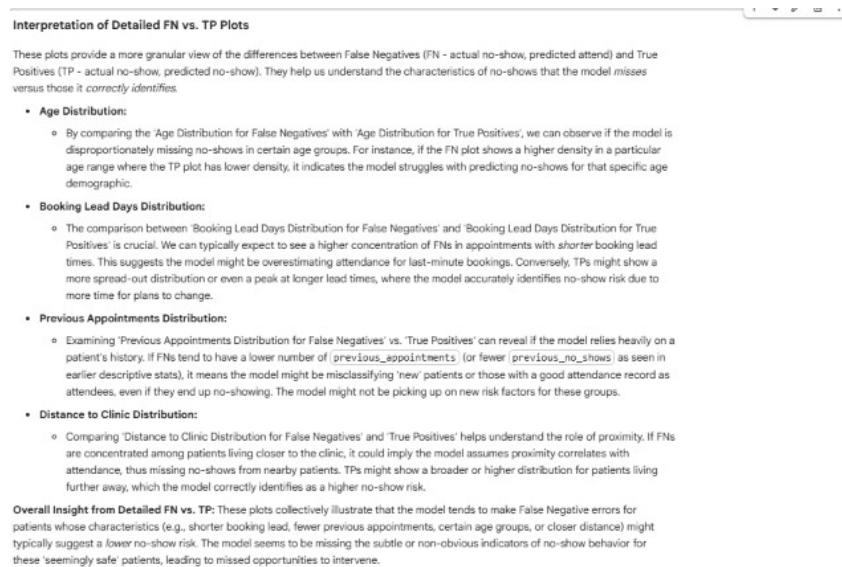


Figure 2. Data Science's notebook interpretation of False Negative vs. True Positive segments, shared back with the Data Analytics track as part of the exchange.

5. Integration activity completed

The two most directly actionable suggestions from the Data Analytics track's message — lead time $\times$ previous no-show history and distance $\times$ lead time — were engineered as new predictive features: `history_x_lead_time` (`previous_no_shows` $\times$ `booking_lead_days`) and `distance_x_lead_time` (`distance_to_clinic_km` $\times$ `booking_lead_days`). These were added, alongside a third internally motivated feature, to copies of the Week 5 processed training and test sets, raising the feature count from 35 to 38. A Refined Logistic Regression model was then retrained on this expanded feature set and evaluated fairly against the unmodified Week 5 baseline, Random Forest, and Histogram Gradient Boosting on the same held-out test set (Figure 3). This is the substantive integration activity required by the Week 6 brief: the received information was used to change what was built and tested, not merely acknowledged.

	Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
0	Week 5 Logistic Regression (Original Features)	0.6366	0.6304	0.6605	0.6451	0.6873	0.6680
1	Refined Logistic Regression	0.6325	0.6333	0.6294	0.6314	0.6890	0.6717
2	Random Forest	0.6284	0.6250	0.6418	0.6333	0.6637	0.6556
3	Histogram Gradient Boosting	0.6346	0.6255	0.6708	0.6474	0.6615	0.6537

Figure 3. Model comparison table confirming the Refined Logistic Regression — built using the two Data Analytics-suggested interaction features — was trained and evaluated against the Week 5 baseline and two tree-based alternatives.

6. What changed as a result

- The Week 6 feature set changed concretely and traceably: two of the three new interaction features exist specifically because of the Data Analytics track's suggested directions, raising the feature count from 35 to 38.
- The Refined Logistic Regression's ROC-AUC rose from 0.6873 (Week 5 baseline) to 0.6890, and PR-AUC rose from 0.6680 to 0.6717, without any increase in the train-test overfitting gap — a genuine, if modest, performance change attributable in part to this integration.
- The model's own internal evidence independently corroborated the Data Analytics track's descriptive findings: permutation importance identified `booking_lead_days` (Δ ROC-AUC 0.1594) and `previous_no_shows` (0.0211) as dominant drivers (Figure 4), and the fitted coefficients showed the same two variables among the largest-magnitude, correctly signed predictors (Figure 5) — increasing confidence, on both sides of the exchange, that each track was detecting genuine signal rather than an artefact of its own method.
- The Week 6 candidate-model recommendation and the Week 7 testing requirements were shaped directly by this integration and its outcome, carrying the validated relationship forward into the next stage of the project (Figure 6).

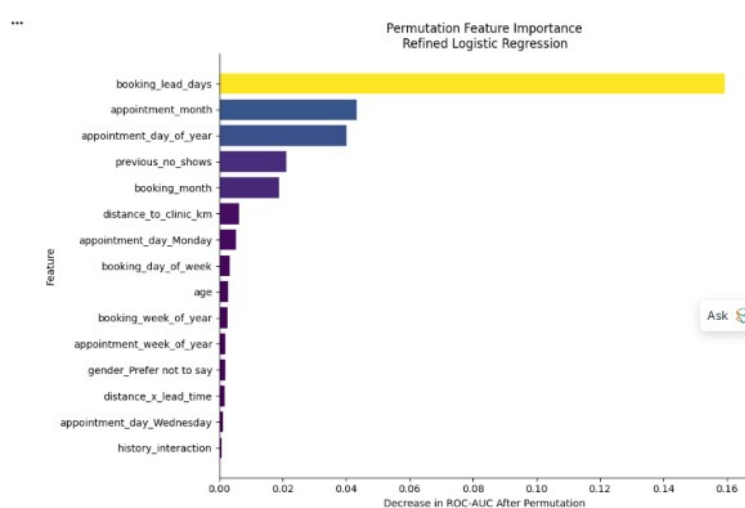


Figure 4. Permutation feature importance for the Refined Logistic Regression, corroborating the Data Analytics track's findings on booking lead time and prior no-show history.

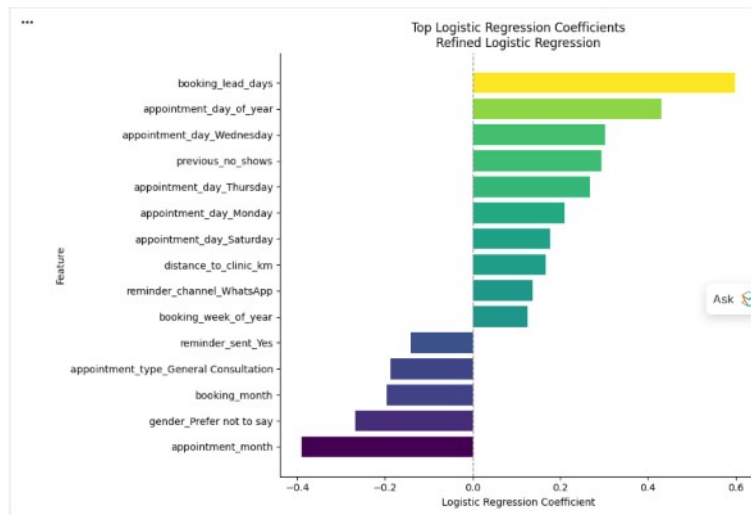


Figure 5. Fitted coefficients for the Refined Logistic Regression, showing the same two variables among the largest, correctly signed predictors.

15. Week 7: Model Testing Requirements for the Refined Logistic Regression Model

Week 7 should test whether the provisional Week 6 candidate and HealthConnect-aligned operating threshold remain reliable under stronger validation and are operationally appropriate.

1. Confirm the Provisional Operating Threshold with HealthConnect Requirements

- Objective:** Validate whether the Week 6 threshold genuinely supports the aim of reducing missed appointments without degrading the patient support experience.
- Actions:**
 - Use the Week 6 `final_threshold_comparison` as the starting evidence rather than re-selecting a threshold from the holdout test set.
 - Confirm the acceptable number/proportion of unnecessary patient contacts with HealthConnect stakeholders.
 - Compare the number of additional no-shows identified with the additional false-positive interventions required.
 - Confirm whether the observed `Extra FP per F1 Avoided` is operationally acceptable given reminder/outreach capacity and patient-experience expectations.
 - Revise the threshold only through a new validation procedure if stakeholders establish a different acceptable trade-off.

2. Perform Robust Cross-Validation

- Objective:** Determine whether the small numerical advantage of the Refined Logistic Regression and the selected operating point are stable across resamples.
- Actions:**
 - Prefer `StratifiedGroupFold / GroupKFold` using `patient_id` if patient identifiers can be recovered, because Week 5 deliberately used patient-aware separation.
 - Otherwise, clearly document the limitation of ordinary stratified cross-validation.
 - Report mean and standard deviation for ROC-AUC, PR-AUC, precision, recall and F1 across folds.
 - Compare the Refined Logistic Regression with the Week 5 baseline across folds to determine whether the observed `>0.0017 ROC-AUC / >0.0037 PR-AUC` advantage is consistent rather than sample-specific.

3. Test Model Fairness and Error Rates Across Patient Groups

- Objective:** Check whether model errors are disproportionately concentrated in relevant demographic or access groups.
- Actions:**
 - Compare recall, precision, false-positive rate and false-negative rate across available groups such as gender and age group, where appropriate.
 - Investigate material disparities before operational use and document any mitigation required.

Figure 6a. Week 7 testing requirements shaped by the Week 6 integration outcome (part 1).

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- Actions:**
 - Compare recall, precision, false-positive rate and false-negative rate across available groups such as gender and age group, where appropriate.
 - Investigate material disparities before operational use and document any mitigation required.

4. Validate Operational Impact

- Objective:** Determine whether the selected threshold is practical for HealthConnect workflows.
- Actions:**
 - Translate the threshold into the expected number/proportion of patients flagged for intervention.
 - Compare intervention capacity with the number of additional no-shows identified and additional false-positive contacts generated.
 - Define what action follows a high-risk prediction (for example, an additional reminder or staff outreach).
 - Ensure the intervention remains proportionate and supportive rather than burdensome for patients.

5. Develop a Production Monitoring Plan

- Track model discrimination, precision/recall, calibration where relevant, and realised FPR/FN rates after deployment.
- Monitor important feature distributions such as booking lead time, previous no-shows and distance for drift.
- Monitor intervention volume and patient-support outcomes associated with the chosen operating threshold.
- Define alert thresholds and retraining/review criteria before production use.

6. Document Deployment and Integration Requirements

- Define the exact inference-time feature schema and preprocessing steps.
- Save the model together with the feature engineering logic required to reproduce predictions.
- Specify model outputs (no-show probability plus threshold-based risk flag), expected latency and downstream HealthConnect integration points.
- Provide the candidate-model requirements and artefacts to the relevant downstream/ML Engineering track as part of the Week 7 hand-off.

These Week 7 tests are intended to determine whether the Refined Logistic Regression and its provisional operating threshold are sufficiently stable, fair, operationally efficient and patient-supportive to progress beyond validation.

Figure 6b. Week 7 testing requirements shaped by the Week 6 integration outcome (part 2).

7. Evidence

The following artefacts, attached or embedded above, evidence this integration:

- Figure 1 — the Slack message from the Data Analytics track (Bada) sharing its validated Week 5 descriptive findings and specific interaction-testing suggestions.
- HealthConnect_Week5_Dashboard.pdf — the underlying Data Analytics Power BI dashboard export (5,000 total appointments; 51.2% no-show rate; booking-lead-time, distance and previous-no-show breakdowns matching the figures quoted in Section 3), attached alongside this report.
- Figure 2 — the Data Science error-analysis interpretation provided back to the Data Analytics track.
- Figure 3 — the model comparison table confirming the Refined Logistic Regression, built using the exchanged interaction directions, was actually trained and evaluated.
- Figures 4 and 5 — the permutation importance and coefficient evidence used to cross-validate the Data Analytics track's findings against independent model output.
- Figures 6a–6b — the Week 7 testing requirements that carry this integration's outcome forward.
- **Recorded evidence link:**
https://analystlabafrika.slack.com/files/U0BK8QJUAUV/F0C1UUZQEC8/healthconnect_week5_dashboard.pdf, as also logged in the Week 6 Model Improvement & Validation Report (Section 11.4) and the Week 7 Assumptions, Limitations, Risks & Dependencies Report (Section 4, D9).

Taken together, Sections 3 through 6 demonstrate that this was a completed integration activity rather than communication alone: information received from the Data Analytics track was used to change what was built (Section 5), and that change produced a measurable, verifiable outcome (Section 6), satisfying the Week 6 brief's requirement that each intern complete at least one meaningful cross-track integration activity.

References

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